Unit 3 — Science and Ethics

Track E · Klasse 12 · Niveau E (Basisfach / Leistungsfach)

Learning objectives I

- *I can read short ethics-of-science texts (CRISPR, large language models, climate intervention) and identify the consequentialist and deontological moves.*
- *I can use vocabulary of science ethics (*precautionary principle, dual use, informed consent, moratorium, governance*).*
- *I can write a 350-word science-ethics essay with a structured ethical argument.*

Lead-in I

Mr. Yilmaz set three short ethics-of-science texts: one on CRISPR germline editing, one on large language models and academic integrity, one on climate intervention research (stratospheric aerosol injection). The class noticed quickly that all three debates rotated around the same question — *who decides what we are not allowed to find out?*

Vocabulary I

- precautionary principle
- dual-use research
- informed consent
- moratorium
- governance
- irreversibility
- externality
- stakeholder engagement
- transparency
- peer review
- replication
- bioethics committee

Vocabulary I

- IRB

Reading I

Reading — *Three Ethics Frames* (paraphrased)
Consequentialism asks *what produces the best outcomes?* — including risks, who bears them, and across what time horizon. *Deontology* asks *what are we required to do regardless of consequences?* — informed consent, dignity, irreversible-harm thresholds. *Virtue ethics* asks *what kind of researcher / institution do we want to be?*. Real-world ethics-of-science debates almost always combine all three; pretending the debate is purely consequentialist is itself a philosophical position.

Vocabulary — science ethics
precautionary principle, dual-use research, informed consent, moratorium, governance,

Reading II

irreversibility, externality, stakeholder engagement, transparency, peer review, replication, bioethics committee, IRB.

Practise I

Niveau E — controlled

1. Match: precautionary principle → caution under uncertainty; dual use → research with both civilian and military potential; moratorium → temporary halt.
2. T or F: consequentialism rules out informed consent; deontology rules out cost-benefit thinking.

Niveau E — productive

Practise II

3. Build 4 sentences applying consequentialist + deontological + virtue-ethics moves to the CRISPR case.

Your turn I

Science-ethics essay, 350 words. Pick one case (CRISPR / LLMs / climate intervention). Build a structured argument with one consequentialist move + one deontological move + one virtue-ethics move + one disagreement / acknowledgement. Use 4 academic discourse markers + 1 cleft.

Sample

The CRISPR-Cas9 germline editing debate is instructive precisely because all three major ethics frames push in slightly different directions, and the resulting policy must be assembled from their overlap. The consequentialist case for some forms of germline editing is, on its own terms, strong: if a single safe intervention can prevent Huntington's disease in a future child, the expected

Your turn II

welfare gain is large and the risk profile, with current CRISPR-Cas9 v3 protocols, is modest. By contrast, the deontological objection is also strong: the future child cannot consent, and the principle of informed consent has, since the Belmont Report (1979), been a non-negotiable in human-subject research. Accordingly, the most useful frame here is the precautionary principle — not the strong version (any risk, no action) but the weak version (irreversible-harm thresholds plus stakeholder engagement). The virtue-ethics question — what kind of researcher community do we want? — tilts toward a moratorium-with-review, partly because the researcher community after the He Jiankui case (2018) has, on reflection, agreed that the breach of governance norms was itself the primary harm. It is precisely this institutional self-correction that makes the moratorium-with-review frame defensible. In this regard, the case is not *for or against germline*

Your turn III

editing but *for or against doing it inside a peer-reviewed, consent-respecting governance structure*. The honest answer is conditional. Some applications, with strong governance, may be permissible; others, without it, are not. The frame is what travels.

Reflect I

- I can identify consequentialist, deontological, and virtue-ethics moves.
- I can use 6+ science-ethics terms.
- I can write a 350-word structured science-ethics essay.

Further reading / listening I

- Henry T. Greely, *CRISPR People* (accessible chapters).
- The Atlantic — *Tech Ethics* essays.